

The even icosahedral representation of conductor 1951: certified finite statements, numerical evidence, and what remains open

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1 October 2026

Abstract

Let ρ be the even icosahedral Artin representation of prime conductor 1951 found by Doud and Moore. Artin’s holomorphy conjecture for ρ is open, and *this paper does not prove it*. We report a computational programme around ρ and say exactly what each result is: a theorem, a theorem modulo named assumptions, a statement conditional on Artin’s conjecture, a floating-point computation, or statistical evidence.

The main points are these. The Frobenius traces of ρ , signs included, are computed from number fields alone (a degree-24 class field fixes the signs), and they agree with the Hecke eigenvalues of a numerically computed Maass form at all 738 primes below 5639. The functional-equation residual of $L(\rho, s)$ is certified to be at most 3.3×10^{-133} , and those of its 37 primitive twists of conductor $q \leq 13$ at most 9.4×10^{-52} ; any pole would need an absurdly small residue. An eigenvalue criterion of Booker–Strömbergsson–Venkatesh type, applied to the automorphised Artin series, gives an odd Maass newform on $\Gamma_0(1951)$ with nebentypus χ whose eigenvalue is within 6.1×10^{-10} of $1/4$ (under stated conditions). Hecke defects $\|(T_p - a_p(\rho))F\|$ for $p = 2, 3$ are bounded over the whole fundamental domain, in floating point, by about 3×10^{-12} . The branch lists behind these bounds carry an exact rational covering certificate, which found and repaired missing branches in four cells. A cached- K_0 , 256-bit interval pass that makes the defect bounds rigorous is built and validated, and 1,316 of its 63,142 records are complete. Assuming $L(\rho, s)$ is entire, it has exactly 553 zeros with $|\operatorname{Im} s| \leq 146$, all simple and on the critical line; for the Galois conjugate ρ' (conductor 1951²) there are 569 with $|\operatorname{Im} s| \leq 96$. Unconditionally, $\zeta_{N_\rho}(s)L(\rho, s)^4$ is entire, where N_ρ is the field cut out by ρ ; $L(\rho, s)$ has an explicit pole-free strip at every height, and a character-theoretic obstruction shows which classes of argument cannot exclude a pole. We also correct two statements made earlier in the project.

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1 Introduction

Artin’s conjecture asserts that $L(\rho, s)$ is entire for every non-trivial irreducible Galois representation ρ . For two-dimensional ρ it is a theorem when the projective image is solvable (Langlands, Tunnell) and when ρ is odd (through Khare–Wintenberger and Serre’s conjecture). The remaining two-dimensional case is even representations with projective image A_5 . Here no single instance is known, and none has been reduced to a finite computation [3, 6, 5].

We study the smallest known example. Doud and Moore [8] found an even icosahedral representation ρ of prime conductor $N = 1951$; Booker, Lee and Strömbergsson [2] show that, assuming Artin’s conjecture, the even non-dihedral representations of conductor at most 2862 are exactly those in their table.

The status ledger. Every result below carries one of five labels:

- [T]: a theorem, with the proof or the exact computation given or cited.
- [T|A]: a theorem modulo explicitly named assumptions (for instance an unrefereed proof or floating-point error bars).
- [C]: conditional on Artin’s conjecture for the L -function concerned.
- [N]: a floating-point computation with a priori error estimates, not interval arithmetic. Such numbers are *not* certificates.
- [S]: statistical evidence.

Table 1 lists the main results. Two principles hold throughout. An enclosure cannot certify an equality, so no computation here can show that an eigenvalue *equals* $1/4$ or that a Hecke eigenvalue *equals* $a_p(\rho)$. And no finite computation can establish holomorphy at every height (Section 13).

2 The representation and its Frobenius traces

Let $N = 1951$ (prime). The least primitive root is 3. Let χ be the even primitive character of order five with $\chi(3) = e(2/5)$ (Conrey label 1951.908). The representation ρ has determinant χ ; twisting by χ^{-3} gives $\rho_0 = \rho \otimes \chi^{-3}$ with $\det \rho_0 = 1$. Its image is $\mathrm{SL}(2, 5)$, the binary icosahedral group, so $\rho = \rho_0 \otimes \chi^3$ and

$$G = \mathrm{Gal}(N_\rho/\mathbb{Q}) \cong \mathrm{SL}(2, 5) \times C_5, \quad |G| = 600.$$

result	status	§
Frobenius traces with signs from number fields; $a_{1951} = \zeta_5^2$; root number	[T]	2
agreement with Maass-form coefficients at all 738 primes $p < 5639$	[N]	3
functional-equation residual $\leq 3.3 \times 10^{-133}$; 37 twists $\leq 9.4 \times 10^{-52}$	[T]	4
odd Maass newform with $ \lambda - \frac{1}{4} \leq 6.1 \times 10^{-10}$	[T A]	5
$\ (T_p - a_p)F\ \lesssim 3 \times 10^{-12}$, $p = 2, 3$, over the whole domain	[N]	6
exact covering certificate for all branch lists; vertex residue $\leq 3.9 \times 10^{-16}$	[T] (geometry)	7
256-bit interval cell bounds; cached- K_0 pass, 1,316 of 63,142 records	[T A] per cell	8
joint quasi-mode: one newform near $1/4$ with $\lambda(2), \lambda(3)$ near a_2, a_3	[N]	9
553 zeros of $L(\rho)$ to height 146; 569 of $L(\rho')$ to height 96	[C]	10
GUE-type spacings; two close pairs certified distinct	[S]/[C]	10
ρ' : conductor 1951^2 , root number, twists	[N]	11
$\zeta_{N_p} L(\rho)^4$ entire; pole-free strip; Brauer formula; obstructions	[T]	12

Table 1: Status ledger of the main results.

Write $a_p = \chi(p)^3 t_p$ with $t_p = \text{tr } \rho_0(\text{Frob}_p) \in \{0, \pm 1, \pm 2, \pm \varphi, \pm(\varphi - 1)\}$, $\varphi = (1 + \sqrt{5})/2$.

Signs from a class field [T]. The A_5 class of Frob_p fixes $|t_p|$, but the sign depends on the lift to $\text{SL}(2, 5)$. It lives in the kernel field N_0 of ρ_0 , a quadratic extension of the A_5 field, unramified at all finite primes and ramified at every real place. Cutting N_0 down by a unipotent subgroup $H \cong C_5$ gives a cyclic quartic extension K_{24}/K_6 of the sextic resolvent

$$K_6 = \mathbb{Q}[y]/(y^6 - 75y^4 - 70y^3 + 826y^2 - 210y - 625), \quad \text{disc} = 1951^4.$$

PARI gives $\text{Cl}(K_6) = C_2$ (certified with `bnfcertify`) and $\text{Cl}^+(K_6) = C_4 \times C_2 \times C_2$. Exactly one index-4 quotient of Cl^+ is cyclic with class field ramified at every real place. Its class field is K_{24} , of degree 24, totally complex, with $\text{nfdisc} = 1951^{16}$. The factorisation of p in K_{24} gives the order of $\rho_0(\text{Frob}_p)$ and hence the sign of t_p . The 2261 primes below 20000 fall into the four predicted rows with counts 27, 575, 769, 890 (Chebotarev: about 38, 565, 754, 904).

The two classes of 5-cycles [T]. Dokchitser resolvents separate the two A_5 classes of 5-cycles. Three independent resolvents agree at all 890 relevant primes below 20000. Which class has $|t| = \varphi - 1$ is fixed by the inertia at 1951: in $\mathbb{Q}_{1951}$, an inertia generator g acts with $\chi(g) = \zeta_5^2$, so $\rho_0(g)$ has eigenvalues ζ_5, ζ_5^4 and trace $\varphi - 1$. The other labelling is the Galois conjugate $\rho' = \rho^\sigma$, $\sigma : \sqrt{5} \mapsto -\sqrt{5}$, which has no inertia-fixed line and conductor 1951^2 (Section 11).

The ramified prime and the root number [T]. The local analysis gives $a_{1951} = \chi(877)^{-3} = \zeta_5^2$, and

$$W(\rho) = -\tau(\chi) \overline{a_{1951}} / \sqrt{1951} = -0.450140208385 - 0.892957889710i.$$

3 Comparison with a Maass form

Against float64 Hejhal coefficients ($M = 8000$) of a Maass form of level 1951, eigenvalue near $1/4$ and nebentypus χ , sign and magnitude agree at all 738 primes $p < 5639$ [N]. This includes the nine

primes with $\text{Frob}_p = +1$ and $p = 5279$, where $\text{Frob}_p = -1$ and $a_p = -2\chi(p)^3$. The agreement is 8×10^{-6} for $p < 1000$ and 2.6×10^{-5} for $p < 2000$. Beyond that the Hejhal coefficients lose accuracy, but the allowed value nearest to the form's coefficient is the Galois one at every prime below 5639. The form's a_{1951} agrees to 7.6×10^{-6} . Its least-squares Fricke constant agrees with $-W(\rho)$ to 1.3×10^{-5} with the Hejhal data, and to 3.8×10^{-15} with the Artin coefficients of Section 5.

4 Functional-equation residuals

For the theta series $A(y) = \sum a_n 4(ny/\sqrt{N})K_0(2\pi ny/\sqrt{N})$ of $L(\rho, s)$, Mellin inversion and Brauer's functional equation give

$$R(y) := A(1/y) - W y \overline{A(y)} = \sum_{\text{poles } s_0} \text{Res}_{s=s_0} \Lambda(s) y^s.$$

This vanishes identically if and only if $L(\rho, s)$ is entire.

Proposition 4.1 ([T], Arb at 1600 bits). *With the exact algebraic a_n , $n \leq 8981$, and the root number above, $|R(y)| \leq 3.3 \times 10^{-133}$ at $y = 0.25, 0.28, \dots, 1.00$, and $|R(y)| \leq 1.3 \times 10^{-132}$ at the reciprocal points. The tail is bounded rigorously using $|a_n| \leq d(n)$ and $K_0(x) < \sqrt{\pi/2x} e^{-x}$.*

Consequently a simple pole on the critical line at height γ would need a residue of at most 5×10^{-128} ($\gamma = 10$), 4×10^{-67} ($\gamma = 100$), 10^{-12} ($\gamma = 180$). The bound degenerates near $\gamma \approx 195$. A sensitivity test [N] shows that the residual responds to every ingredient: the wrong 5-class labelling gives 1.3, flipping the sign of a_2, a_3 or a_{13} gives 1.4–1.7, and conjugating W gives 1.8.

Twists. For primitive $\psi \bmod q$, $(q, 1951) = 1$, the twist $\rho \otimes \psi$ has conductor $1951q^2$ and root number $W_\psi = W(\rho)\psi(-1)\chi(q)\psi(1951)\tau(\psi)^2/q$. In float64, all 661 primitive ψ with $q \leq 60$ fit this root number to 1.7×10^{-14} [N]. In Arb, all 37 primitive twists with $q \leq 13$ satisfy $|R_\psi(y)| \leq 9.4 \times 10^{-52}$ at five points [T]. A pole on the line would need a residue of at most 8×10^{-47} at height 10 and 3×10^{-6} at height 70.

Remark 4.2. None of this proves that $L(\rho, s)$ is entire. There is no lower bound for the residue of a pole, several poles could in principle cancel at every sample point, and by [3] a non-automorphic ρ would have infinitely many poles. The residuals test the conductor and root number, and they rule out large, isolated, low poles. They cannot rule out anything at every height.

5 An odd Maass newform near eigenvalue $1/4$

The companion note [9] proves a version of the Booker–Strömbergsson–Venkatesh criterion [7] for $\Gamma_0(N)$ with a primitive nebentypus. We recall its form. Let $\mathfrak{F}^+$ be the Ford domain of $\Gamma^+ = \langle \Gamma_0(1951), W_N \rangle$: 921 non-vertical sides, 922 vertices, 299 vertex cycles, volume $\frac{\pi}{6}(N+1)$, and lowest vertex at height exactly $\sqrt{3}/(2N)$. All of this is computed exactly. Given an odd finite Fourier series

$$\tilde{f}(z) = \sum_{0 < |n| \leq M} c_n \sqrt{y} K_0(2\pi|n|y) e(nx),$$

whose terms are all Δ -eigenfunctions with eigenvalue $1/4$, define $F = \tilde{f}$ on $\mathfrak{F}^+$, $F(Wu) = \tilde{\lambda} \overline{\tilde{f}(\iota u)}$, and extend by χ -automorphy. The criterion bounds the distance from $1/4$ to the odd cuspidal spectrum by the L^2 norms Q_0, Q_1 of the jumps of F and of its normal derivative across the 921 sides. Odd forms have no continuous spectrum, and Green's identity reduces a two-dimensional supremum to one-dimensional integrals.

Theorem 5.1 ([T|A]; [9]). *There is an odd Maass cusp newform ϕ on $\Gamma_0(1951)$ with nebentypus χ whose eigenvalue satisfies $|\lambda_\phi - \frac{1}{4}| \leq 5.33 \times 10^{-5}$ (Hejhal coefficients, $M = 8000$). With the exact Artin coefficients $a_n(\rho)$, $n \leq 12000$, as c_n and $\tilde{\lambda} = -W(\rho)$, the same criterion gives*

$$\left| \lambda_\phi - \frac{1}{4} \right| \leq 6.1 \times 10^{-10}.$$

Assumptions. The proof of the criterion in [9] has not been refereed. The jump norms $Q_0 \leq 4.43 \times 10^{-21}$ and $Q_1 \leq 1.25 \times 10^{-20}$ are computed in floating point with a priori error bars by two independent programs, and the larger is used; they are not interval enclosures. Theorem 5.1 does not say that $\lambda_\phi = \frac{1}{4}$, nor that ϕ corresponds to ρ .

6 Hecke defects over the whole fundamental domain

For $p \neq 1951$ let

$$(T_p F)(z) = \frac{1}{\sqrt{p}} \left(\chi(p) F(pz) + \sum_{b=0}^{p-1} F\left(\frac{z+b}{p}\right) \right), \quad D = (T_p - a_p)F = \sum_{i=0}^{p+1} w_i F(G_i z),$$

with weights $w = (-a_p, \chi(p)/\sqrt{p}, 1/\sqrt{p}, \dots, 1/\sqrt{p})$. Here F is the automorphised Artin series of Section 5. The full Artin series is a T_p -eigenfunction by algebra, so D measures truncation, inconsistency across the seams, and rounding.

The domain. The fundamental domain $\mathfrak{F}^+ \cup W\mathfrak{F}^+$ of $\Gamma_0(1951)$ is cut into cells in $(x, \log y)$ below $y = p/\sqrt{N}$. Above that height every Hecke point lies in the strip $y > 1/\sqrt{N}$, where $F = \tilde{f}$ and D is a pure Fourier truncation tail, of size roughly e^{-1700} . That tail is not part of the cell computation.

Branches and value maps. For each cell and each Hecke point $G_i z$, the word M that reduces $G_i z$ into $\mathfrak{F}^+$ changes only where $G_i z$ crosses an isometric circle, that is where z crosses a preimage geodesic. Branches are detected on the geodesics in 40-digit arithmetic. Each word comes with a value map $F(G_i z) = \alpha \tilde{f}(Mz) + \beta \tilde{\bar{f}}(Mz)$ in which exactly one of α, β is non-zero, of modulus one: a fifth root of unity, or $-\Lambda$ times one, with $\Lambda = \tau(\chi)\zeta_5^3/\sqrt{1951}$. The cell bound is the bound for the base formula plus, for each Hecke point, the largest bound over its other branches of the difference.

Interpolation (method v3). The map $u \mapsto \tilde{f}(Mu)$ extends analytically to $\{|\operatorname{Im} x| < \operatorname{Re} y\}$ in the complexified coordinates, for every real M with positive determinant. Termwise,

$$|\tilde{f}(X + iY)| \leq \sqrt{|Y|/\operatorname{Re} Y} S(\operatorname{Re} Y - |\operatorname{Im} X|), \quad S(g) = \sum_{n \geq 1} |a_n| n^{-1/2} e^{-2\pi n g}, \quad (1)$$

from $|K_0(Y)| \leq K_0(\operatorname{Re} Y) < \sqrt{\pi/2 \operatorname{Re} Y} e^{-\operatorname{Re} Y}$. This gives closed-form Bernstein constants on polyellipses. $\tilde{f}$ is tabulated on tensor Chebyshev grids in image space $(X, \log Y)$, where the Bessel and sine factors separate:

$$\tilde{f}(X_a + iY_b) = 2i\sqrt{Y_b} \sum_n a_n K_0(2\pi n Y_b) \sin(2\pi n X_a)$$

is one matrix product. It is then interpolated barycentrically to the images of the cell's Chebyshev nodes. The cell bound is $\Lambda_{n_x} \Lambda_{n_l} (\max_{\text{nodes}} |D| + E) + 8B(r_x^{-n_x} + r_l^{-n_l})/((1 - r_x^{-1})(1 - r_l^{-1}))$, with $\Lambda_n \leq \frac{2}{\pi} \log(n+1) + 1$ for first-kind Chebyshev nodes [13].

The Fricke identity. For the automorphised F one has $D(Wz) = \eta\chi(p)\overline{D(z)}$ with a constant $|\eta| = 1$, so $|D(Wz)| = |D(z)|$. This was checked on a non-automorphic control series, where $|D| \sim 0.2\text{--}0.9$, to 3×10^{-16} . The I - and W -jobs of an initial cell therefore bound the same function, and $2 \sum \min(S_I, S_W)$ is also a valid bound.

Results [N].

	jobs	final cells	worst cell	$\sqrt{\sum \text{area} \cdot \text{sup}^2}$	Fricke form
T_2 (fresh run, current code)	18,512	30,615	1.658×10^{-13}	2.86373×10^{-12}	2.37757×10^{-12}
T_3	18,598	32,527	1.961×10^{-13}	3.52391×10^{-12}	2.81698×10^{-12}

The T_2 figure comes from a rerun with the current branch code. An earlier 8-worker run, made before a fix to the branch finder, gave 2.8635×10^{-12} ; its checkpoint is superseded, because covering it with the current code would certify lists other than the ones bounded. These cell bounds are floating-point bounds. The 256-bit recomputation (Section 8) shows they are dominated by rounding.

7 An exact covering certificate for the branch lists

The cell bound of Section 6 bounds $\sup_C |D|$ only if the branch list *covers* the cell: every $u \in C \cap \overline{\mathfrak{F}^+}$ must have some listed word M with Mu in a translate of $\overline{\mathfrak{F}^+}$. The branch finder is a 40-digit heuristic, so this needs a separate proof.

Method [T]. A condition “ Mu lies outside disk $j + m$ ” is $q(u) = A|u|^2 + Bx + C \geq 0$ with exact rational A, B, C , and its exact minimum and maximum over a dyadic box are elementary. A box is covered when, for every sign pattern of its undecided conditions, some listed word (or the pseudo-branch “ $u \notin \mathfrak{F}^+$ ”) has all its conditions satisfied. Conditions on a shared circle are identified exactly as $\pm$ multiples. Uncovered boxes are subdivided to side 2^{-100} , with a budget of 10^5 boxes. A record that reaches the budget is *not certified*. Further details:

- The image box is a rigorous Arb enclosure: the exact union of the rational corners, rounded outward.
- The circle pre-filter has margin $0.1r \geq 4.8 \times 10^{-5}$, against float error of about 10^{-15} .
- Every integer translate whose disks can meet the image box is examined. Because $\tilde{f}$ is 1-periodic and translations have trivial value map, “outside every disk of every translate” is then exactly the right condition, and no strip predicate is needed.
- Branch repair is a search, not part of the proof. From a failing point, the 40-digit reduction proposes an integer word, and only the exact cover that follows certifies anything. These words are integer matrices of determinant $p \cdot 1951^k$ (the Hecke step times k Fricke steps), not elements of $\text{SL}_2(\mathbb{Z})$; the certifier uses them only as linear fractional maps.

Theorem 7.1 ([T], exact rational arithmetic). *Every one of the 30,615 final T_2 records and 32,527 final T_3 records is covered by its branch list, after repair, except for leftover boxes of total Euclidean area 1.49×10^{-42} (T_2) and 4.09×10^{-42} (T_3). No record reached the budget. The largest number of boxes used was 4,989 (T_2) and 6,077 (T_3). For T_2 , no words had to be added. For T_3 , seven records in four cells needed added words:*

cell x -range	A , point	listed $\rightarrow$ repaired	float bound, listed $\rightarrow$ repaired
$[-0.40686, -0.40653]$	$I, 2$	$2 \rightarrow 4$	$3.659 \rightarrow 3.683 \times 10^{-14}$
$[-0.16102, -0.16049]$	$W, 1$	$2 \rightarrow 4$	1.260×10^{-13} (unchanged)
$[-0.16102, -0.16049]$	$I, 2$	$2 \rightarrow 4$	7.207×10^{-14} (unchanged)
$[-0.14892, -0.14887]$	$I, 3$	$3 \rightarrow 4$	3.777×10^{-14} (unchanged)
$[-0.14892, -0.14887]$	$W, 4$	$3 \rightarrow 4$	6.255×10^{-14} (unchanged)
$[0.16049, 0.16102]$	$W, 1$	$2 \rightarrow 5$	$1.234 \rightarrow 1.265 \times 10^{-13}$
$[0.16049, 0.16102]$	$I, 2$	$2 \rightarrow 5$	6.993×10^{-14} (unchanged)

All 14 added words have determinant $3 \cdot 1951^k$, $k \in \{2, \dots, 6\}$. With the repaired lists, the plain T_3 figure moves from 3.52391×10^{-12} to 3.52392×10^{-12} .

On the repaired records the old number was not a bound. Pointwise, the listed words left 39% of the points of the cell at $x \approx -0.4069$ uncovered, and 6% at $x \approx 0.1605$. The added formulas agree with the listed ones to rounding, as automorphy of the form predicts, so the bounds barely move.

Proposition 7.2 (vertex residue, $[T]$, Arb). *On $\overline{\mathfrak{F}}^+$, $\text{Im } w \geq y_{\min} = \sqrt{3}/3902$ exactly: the 921 arcs of the floor cover $[-\frac{1}{2}, \frac{1}{2}]$ contiguously, each is concave, and the minimum over each arc is at its rational endpoints. Hence for every u ,*

$$|D(u)| \leq \left(\sum_i |w_i| \right) S(y_{\min}) \leq \begin{cases} 3.12133 \times 21.69512 \leq 67.718 & (p = 2), \\ 3.92744 \times 21.69512 \leq 85.2062 & (p = 3), \end{cases}$$

The leftover boxes (the vertex residue) lie at height $y \geq y_{\min}$, so their hyperbolic area is at most their Euclidean area divided by $y_{\min}^2$: at most $1.49 \times 10^{-42} \cdot (3902^2/3) \leq 7.6 \times 10^{-36}$ (T_2) and $4.09 \times 10^{-42} \cdot (3902^2/3) \leq 2.1 \times 10^{-35}$ (T_3). They therefore add at most $\sqrt{67.718^2 \cdot 7.6 \times 10^{-36}} \leq 1.9 \times 10^{-16}$ (T_2) and $\sqrt{85.2062^2 \cdot 2.1 \times 10^{-35}} \leq 3.9 \times 10^{-16}$ (T_3) to $\|(T_p - a_p)F\|$. The factor $1/y_{\min}^2 \approx 5 \times 10^6$ from the measure $dx dy/y^2$ is what lifts this from the 10^{-19} that the Euclidean area alone would suggest.

Remark 7.3 (a correction). A first version of the certifier was reported to have found three missing-branch cells in a 5,375-record T_3 snapshot, with leftover vertex area of about 10^{-43} . Grok’s reading of the output file was correct on both counts. The real leftover in that snapshot was 1.23×10^{-7} , coming entirely from records that had hit a 4,000-box cap. Two of the three “repairs” were artifacts of that cap: pending boxes had been counted as uncovered. The version above fixes this, encloses image boxes rigorously, examines all translates, and supersedes the earlier files. The earlier figure “ $< 10^{-20}$ ” for the vertex residue was also wrong; Proposition 7.2 gives the correct order, 10^{-16} .

8 Interval arithmetic for the cell bounds

Per-cell interval bounds $[T|A]$. In `interval_cell.py` every quantity in the cell bound is an Arb ball at 256 bits:

- the exact a_n , a_p , $\chi(p)$ and $1/\sqrt{p}$;
- value maps snapped to their exact algebraic constants;
- exact first-kind Chebyshev nodes and their images;
- K_0 and $\sin$ in Arb;

- the majorant S summed in Arb and evaluated at g 's lower end;
- every interpolated image point asserted to lie inside $[-1, 1]^2$.

On nine T_2 cells from the top of the domain to the floor, and on six further cells (among them the elliptic-vertex cell and the three worst floor cells of the float run), the interval bounds lie between 1.97×10^{-16} and 1.34×10^{-15} . That is 55–120 times below the float bounds. The node values of D enclose to about 10^{-21} , so the float figures of Section 6 were rounding noise, and the interval bounds are set by the interpolation tolerances, not by D . The assumptions are those of Section 7 for the geometry, the cited Lebesgue-constant inequality, and, for the whole-domain statement, the cusp tail.

Cached- K_0 design. One cell of this code took about 480 s, 467 of them in the $\tilde{f}$ -table. A 256-bit K_0 costs about 114 μ s, and every map rebuilt its own grid. `interval_cell12.py` keeps every term of the bound and changes only the tabulation:

1. *Global log- Y grid.* Scales $h \in \{\frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1\}$, intervals $[kd - h, kd + h]$ with $d = h/2$, and fixed node counts 40, 48, 72, 128. Each map uses the smallest interval containing its image range. The table $a_n K_0(2\pi n Y_b)$ is built once per interval and reused. A simulation over 1,500 records finds 115 (T_2) and 121 (T_3) distinct tables and no fallbacks.
2. *Large arguments.* For $x > 50$ the table entry is the ball $[0 \pm \sqrt{\pi/2x} e^{-x}]$, valid because $0 < K_0(x) < \sqrt{\pi/2x} e^{-x}$; the majorant is taken at x 's lower end. At 256 bits Arb's own K_0 is very loose at large argument (its radius is 38 times the value at $x = 100$). Below 50 its relative radius is at most 10^{-31} .
3. *Exact integer products.* SK is computed as an exact FLINT integer product of fixed-point matrices at scale 2^{-112} . The sines come from an exact integer rotation $w_n = \lfloor w_{n-1} \tilde{z} / 2^{112} \rfloor$ with $e_n \leq n(1 + \delta_0)^n (\delta_0 + \sqrt{2} \cdot 2^{-112})$. Column b is enclosed by $\varepsilon_S \sum_n |K_{nb}| + (1 + \varepsilon_S) \sum_n |K_{nb} - \tilde{K}_{nb}|$.

Validation.

- The fixed-point ball contains the Arb product, and a 2^{-94} corruption is detected.
- On the T_2 floor cell, the per-map method gives 9.81×10^{-16} . The final cached code gives 8.98779×10^{-16} ; earlier versions gave 8.98654 and 8.98621×10^{-16} . Each is an upper bound; they are different balls, not a bit-stable constant. The float bound is 8.475×10^{-14} .
- Node by node, on all maps of a top T_2 cell, the T_2 floor cell and a random T_3 record, cached and per-map values agree to about 2×10^{-21} , inside combined error bounds of about 10^{-17} .
- The repaired T_3 record at $x \approx 0.1605$ (W) uses its repaired list and gives 1.02×10^{-15} .

Status of the pass. 1,316 records are complete (1,116 for T_2 , 200 for T_3), with 0 errors and 0 fallbacks. The interval/float ratio has median 0.010 (T_2) and 0.009 (T_3). Measured steady rates are about 11 s per record at the T_2 floor, 3–6 s at $y \approx 10^{-3}$, and about 10 s for T_3 at $y \approx 10^{-3}$, plus about 1.2 h of tables per worker. These project to about 4.5–5 h (T_2) and 6.5–8 h (T_3) on 16 cores. That is a projection, not a measured full run. A summary is a domain bound only when all records are present. The partial sums are not bounds.

9 A joint quasi-mode

Lemma 9.1 ([T]). *Let $\{\phi_j\}$ be an orthonormal basis of the odd χ -space of level 1951 consisting of joint eigenfunctions of Δ and the T_p , $p \neq 1951$. Suppose u is in the domain of Δ with $\|(\Delta - \frac{1}{4})u\| \leq \varepsilon_0 \|u\|$ and $\|(T_p - a_p)u\| \leq \varepsilon_p \|u\|$ for $p \in S$. Then some ϕ_j satisfies $|\lambda_j - \frac{1}{4}| \leq \sqrt{2} \varepsilon_0$ and $|\lambda_j(p) - a_p| \leq \sqrt{2|S|} \varepsilon_p$ for every $p \in S$.*

Proof. Put $w_j = |\langle u, \phi_j \rangle|^2 / \|u\|^2$, so $\sum w_j = 1$. Then

$$\sum_j w_j \left(\frac{|\lambda_j - \frac{1}{4}|^2}{\varepsilon_0^2} + \sum_{p \in S} \frac{|\lambda_j(p) - a_p|^2}{|S| \varepsilon_p^2} \right) \leq 2,$$

so some j has the bracket at most 2. □

Take $u = F_S = k * F$, the smoothed function of the eigenvalue criterion. Convolution commutes with T_p , and $\|k * g\| \leq (\int k) \|g\|$, which gives $\varepsilon_p \leq \sqrt{1 + 1/Y} \|(T_p - a_p)F\| / \|\tilde{f}\|_{Y_1}$ with $Y = 6.7869$ at $\delta = 0.75$ and $\|\tilde{f}\|_{Y_1} \geq 1.71$.

The same function must be used (a correction). u must be built from the *same* F as the eigenvalue certificate, F_{cert} : exact Artin coefficients, $n \leq 12000$. The float defect runs of Section 6 bound a different function, F_{run} : float64 coefficients, $n \leq 60000$. Since $\|T_p\| \leq (p+1)/\sqrt{p}$,

$$\|(T_p - a_p)F_{\text{cert}}\| \leq \|(T_p - a_p)F_{\text{run}}\| + \left(|a_p| + \frac{p+1}{\sqrt{p}}\right)(\|\Delta F_{\text{trunc}}\| + \|\Delta F_{\text{round}}\|),$$

where both differences are bounded through (1), height by height, over $\mathfrak{F}^+ \cup W\mathfrak{F}^+[\mathbf{N}]$:

- the truncation difference ($12000 < n \leq 60000$) satisfies $\|\Delta F_{\text{trunc}}\| \leq 4.17 \times 10^{-14}$;
- the rounding difference, using the measured per-coefficient errors $|a_n^{\text{float}} - a_n| \leq 3.6 \times 10^{-14}$ (median 0) with a safety factor 2, satisfies $\|\Delta F_{\text{round}}\| \leq 2.34 \times 10^{-12}$.

The rounding term is of the same size as the defect bounds, so it cannot be dropped.

Corollary 9.2 ([N]; float-level). *Under the assumptions of Theorem 5.1 and with the floating-point cell bounds, there is an odd Hecke–Maass newform ϕ of level 1951 and nebentypus χ with*

$$\left| \lambda_\phi - \frac{1}{4} \right| \leq 8.7 \times 10^{-10}, \quad |\lambda_\phi(2) - a_2(\rho)| \leq 1.3 \times 10^{-11}, \quad |\lambda_\phi(3) - a_3(\rho)| \leq 1.7 \times 10^{-11}.$$

Project notes stated 3.6×10^{-12} for the last two bounds. Those numbers combined bounds for two different functions and are superseded by Corollary 9.2. Once the interval pass of Section 8 is complete, it bounds the exact-coefficient series and the rounding term disappears. Inserting the *float* defect bounds for that series into the same argument gives a float estimate of about 3.8×10^{-12} and 4.7×10^{-12} . This is a projection, not a measured interval bound. The finished interval cell bounds have been about 100 times smaller than the float ones (Section 8), but the actual figures will only be known when the pass is complete. In every form this is a finite statement about one form at two primes. It is not $\lambda_\phi = \frac{1}{4}$, and it is not $\lambda_\phi(p) = a_p(\rho)$.

10 Zeros, conditional on entireness

$Z(t) = W^{-1/2}\Lambda(\frac{1}{2} + it)$ is computed in Arb from the full-line theta integral by the trapezoid rule, with error bound $2M/(e^{2\pi a/h} - 1)$ [14]. The majorant M is bounded using $|a_n| \leq d(n)$ and $|K_0(w)| \leq K_0(\operatorname{Re} w)$. The count up to height T_0 uses Theorem 2.1 of Palojärvi–Zhao [12], with all constants as Arb balls.

Theorem 10.1 ([C]). *1. If $L(\rho, s)$ is entire, it has exactly 277 zeros with $0 < \operatorname{Im} s \leq 146$ and 276 with $-146 \leq \operatorname{Im} s < 0$. All are simple and on the critical line. (15,001 certified signs; Turing margins +1.86 and +1.72.)*

2. If $L(\rho', s)$ is entire, it has exactly $285 + 284 = 569$ zeros with $|\operatorname{Im} s| \leq 96$, all simple and on the critical line. (10,001 certified signs; margins +1.01 and +1.48; the 313 intervals below height 56 reproduce an earlier run.)

The condition enters twice: through the functional equation for $u < 0$, and through [12], which needs an entire L -function. Simplicity follows because the argument principle counts with multiplicity and equals the number of sign-change cells.

Spacing statistics [S]. After unfolding, the Kolmogorov–Smirnov distance of the nearest-neighbour spacings to the Wigner surmise is 0.038 (ρ) and 0.032 (ρ'). Against Poisson it is 0.310 and 0.281; against two superposed independent GUE spectra, 0.175 and 0.147. The 5% critical value is about 0.082. The Wigner surmise only approximates the Gaudin law, and about 280 spacings is a small sample. The cross-statistic of ρ' zeros against ρ gaps (KS 0.057 to uniform) is consistent with independence. The two closest pairs are certified distinct [C]: at $t \approx 38.432$ the gap is at least 3×10^{-4} , and at $t \approx 64.043$ it is at least 5×10^{-4} . This is the behaviour expected of two distinct primitive degree-2 L -functions. It proves nothing about poles.

11 The Galois conjugate ρ'

$\rho' = \rho^\sigma$ has the same determinant χ . Tame local Langlands gives $\rho'|_{I_{1951}} = \chi^2 \oplus \chi^4$, so its conductor is 1951^2 , $a_{1951^k} = 0$, and $W(\rho') = -g(\chi^2)g(\chi^4)$, with g the normalised Gauss sum. In float64 [N]:

- The theta relation holds at $q = 1951^2$ with $|W| = 1$ to 12 digits and residual 2.4×10^{-15} . It fails at $q = 1951$ and $q = 1951^3$, with residuals 0.42 and 0.26. The fitted W equals $-g(\chi^2)g(\chi^4)$ to 10^{-15} .
- For the two cubic characters mod 7 and the four order-5 characters mod 11, all twelve twists of ρ and ρ' satisfy the theta relation at the predicted conductor, with $W = W(\pi)\psi(N_\pi)\chi(q)g(\psi)^2$ to 1.4×10^{-14} . The neighbouring conductors fail.

These are checks of conductor and root number. Every Artin L -function has a functional equation by Brauer, so they are not evidence of automorphy as such.

12 Unconditional structural results

Proposition 12.1 ([T]). $\zeta_{N_\rho}(s) L(\rho, s)^4$ is entire. Precisely,

$$\zeta_{N_\rho}(s) L(\rho, s)^4 = \zeta_{N_0}(s) \prod_{j=1,2,4} L_{N_0}(\chi^j \circ \operatorname{Nm}, s) \cdot L_M(\chi^3 \circ \operatorname{Nm}, s) \cdot L_{K_{12}}(\lambda \otimes \chi^3 \circ \operatorname{Nm}, s)^4 \cdot L_{K_5}(\rho|_{K_5}, s)^2,$$

a product of Hecke L -functions and a tetrahedral L -function over the quintic field (entire by Langlands–Tunnell). So a pole of $L(\rho, s)$ of order m is a zero of ζ_{N_p} of order at least $4m$. The exponent 4 is optimal among Heilbronn-type arguments: an integer program over the 165 inequalities from all Hecke and Langlands–Tunnell L -functions of G has optimum 4, with a dual certificate.

Corollary 12.2 (pole-free strip, [T]). *Using Lee’s zero-free region [11] with $n = 600$ and $\log d_{N_p} = 480 \log 1951$: $L(\rho, \sigma + it)$ has no pole for $\sigma \geq 1 - 1/(44548 + 5721 \log |t|)$, $|t| \geq 1$, nor, by the functional equation, in the mirror strip.*

The strip does not reach the critical line, and cannot by this method.

A Brauer formula and an obstruction [T]. $L(\rho, s) = L_{K_{12}}(s)/L_{K_{10}}(s)$, a ratio of Hecke L -functions of degrees 12 and 10. So every pole of $L(\rho)$ lies at a zero of the degree-10 function. The virtual character $\theta^* = (6 \otimes \chi^3) - \rho$ restricts to a genuine character on every subgroup of G not containing $\mathrm{SL}(2, 5)$. Hence no argument that uses only holomorphy of L -functions induced from proper subfields (other than $\mathbb{Q}$ and the quintic field), or of Artin L -functions of irreducibles other than $\rho, \bar{\rho}$, can exclude a pole of $L(\rho)$. The full cone of locally genuine virtual characters has 15 extreme rays, 10 of them obstructions. One of them pairs non-negatively with every σ -invariant character, so a proof must break the symmetry $\rho \leftrightarrow \rho'$. By contrast, the p -adic Artin conjecture holds for ρ at every odd p [10, 15].

13 What remains open, and what would close it

Artin’s conjecture for ρ remains open. A proof needs one of two things:

- a construction of Galois representations from Maass forms with eigenvalue $1/4$;
- holomorphy of $L(\rho, s)$ at every height.

The first is open: such forms have no known cohomological realisation, and the symmetric-power route stops before the icosahedral case; Booker [5] states that his method cannot recover the icosahedral Galois representation. The second is a statement about infinitely many heights; by [3] it would suffice to show that there are only finitely many poles, which is still not a finite statement. Booker’s finite-height method [4] needs entire L -functions that contain ρ ; at conductor 1951 the smallest has degree 10 and conductor about 2×10^{26} , out of reach. Every computation here is finite. A finite computation gives enclosures, which cannot certify equalities, and statements up to a height. So the programme cannot by itself prove Artin for ρ .

Next steps, each a finite statement.

1. Complete the cached- K_0 interval pass for T_2 and T_3 . With the covering certificate and the vertex residue, it gives rigorous defect bounds for the exact-coefficient series, modulo the circle list and pairing matrices, the Lebesgue inequality and the cusp tail.
2. Make the eigenvalue certificate rigorous, with interval jump norms, and use one function for every bound.
3. Isolation near $1/4$: a trace-formula count showing a unique odd newform in a window, which would identify the forms of Sections 5 and 9. The gap to the next eigenvalue is unknown. A heuristic Weyl count suggests other eigenvalues within a few hundredths of $1/4$.
4. Certified enclosures of Maass-form coefficients, to turn Section 3 into a theorem prime by prime.

Acknowledgements

Several of the corrections recorded here were found by an independent reader, the model Grok, who checked the output files line by line: the covering-snapshot leftover, the cap artifacts, the code version of the T_2 checkpoint, and the stability wording. The computations used PARI/GP, FLINT/Arb through python-flint [1], NumPy and mpmath.

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